



HK IMO Training 2021-2022
Problem Set 27
For 30 April 2022



Problem 105. Let x be a positive real number, and let n be a positive integer. Prove that

$$x^n + \frac{1}{x^n} - 2 \geq n^2 \left(x + \frac{1}{x} - 2 \right).$$

Problem 106. Given $\triangle ABC$, define the following points.

- D is the intersection point of the internal angle bisector of $\angle BAC$ and the side BC
- E is the intersection point of the tangent at A to the circumcircle of $\triangle ABC$ and the line BC
- F is the point on the A -altitude of $\triangle ABC$ such that $DF \perp AE$
- M is the midpoint of AB
- P is the intersection point of AB and EF
- Q is the intersection point of AD and EM
- O is the circumcentre of $\triangle ABC$

Prove that $PQ \perp AO$.

Problem 107. Find all prime numbers p and all odd positive integers m and n such that $m^4 + 4^n p^4$ divides $m^2(m^4 - 4^n p^4)$.

Problem 108. Let k and n be positive integers. We play a game on an infinite row of boxes as follows. Initially, we distribute n balls to the boxes in some ways, where each box may contain more than 1 ball. In each operation, we are allowed to make one of the following moves:

- We choose two adjacent nonempty boxes, and transfer all the balls from one box to the other.
- We choose a box with at least 2 balls. Then we transfer one ball to the k th box on the left, and transfer one ball to the k th box on the right.

The game ends when we cannot make any move.

- (a) Prove that the game must end if $n \leq k + 1$.
- (b) Determine all k such that there exists an n for which the game may never come to an end.